

Mechanisms of interactions between soil minerals and dissolvable biochar components; [■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■]

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ABSTRACT ORIGINAL

The interactions between dissolvable biochar components and soil minerals were explored from the perspective of minerals to reveal the mechanism of biochar stability in soil. Batch experiments on the adsorption of dissolvable biochar components by soil minerals were carried out. Biochar was prepared from rice straw under different carbonization temperatures (RS300, RS500, and RS700). Kaolinite, montmorillonite, and illite were selected as three representative soil minerals. The results showed that the adsorption capacity of soil minerals gradually increased as the carbon concentration of dissolvable biochar components increased. The order of adsorption capacity of the three soil minerals, montmorillonite > illite > kaolinite, was related directly to their structures. The proportions by which each mechanism contributed to the stability of the representative soil minerals also differed. The binding mechanisms of both kaolinite and montmorillonite to the dissolvable biochar components, whose contribution proportions were 3.4%~87.0% and 32.0%~82.0%, respectively, were a result of the van der Waals force. The binding of illite to the dissolvable components of RS300 was primarily conducted by Ca^{2+} , at a contribution ratio of 60.4%~70.6%; its binding to RS500 and RS700, with contribution ratios of 18.7%~65.0% and 53.0%~67.6%, respectively, was a result of the van der Waals force. Comprehensive comparative analysis shows that RS500 can inhibit the dissolution of dissolvable components to the greatest extent by interactive combination with montmorillonite, which is conducive to utilizing the carbon sequestration and emissions reduction advantages of biochar. © 2022 Editorial Board of Journal of Agro-Environment Science. All rights reserved.